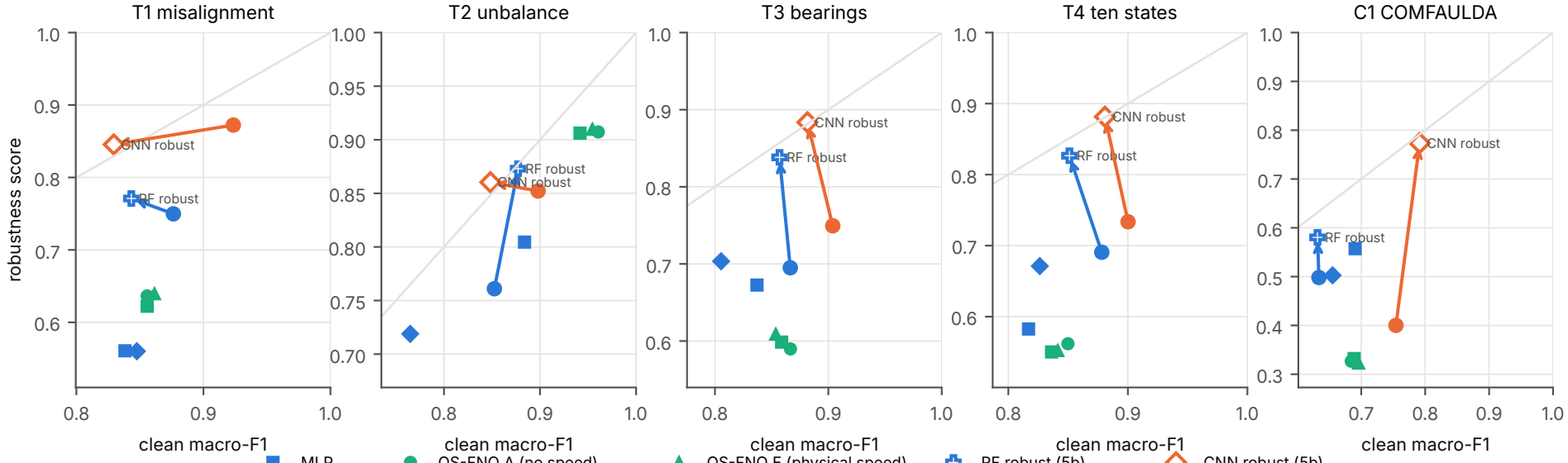
Clean accuracy vs robustness (mean over ±1/2/5 % speed error and dead sensor); arrows: parent → robust training
T1 misalignment
T2 unbalance
T3 bearings
T4 ten states
C1 COMFAULDA
robustness score
clean macro-F1
CNN robust
RF robust
MLP
XGBoost
OS-FNO A (no speed)
OS-FNO B (speed in head)
OS-FNO E (physical speed)
RF
RF robust (5b)
CNN (order spectra)
CNN robust (5b)